

Homework Assignment 3

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All team members contributed in equal measure

Instrumentation: Reported Node Counts

The following counts represent the number of times `inc_count/0` was called (once per evaluated board) during the execution of `minimax_value/3` for the given positions.

Position Description	Board State (Row-Major)	Player to Move	Minimax Value	Nodes Evaluated
Empty Board	[e,e,e, e,e,e, e,e,e]	x	0 (Draw)	549,946
Mid-Game	[x,e,e, e,o,e, e,e,x]	o	0 (Draw)	1053
Late Game	x,o,x, o,x,o, e,e,e]	x	1 (X Wins)	8

Verification Commands

To reproduce these counts, you can run the following queries in the Prolog REPL:

1. Empty Board:
`clear_count, minimax_value([e,e,e,e,e,e,e,e,e], x, V), get_count(N).`
2. Mid-Game:
`clear_count, minimax_value([x,e,e,e,o,e,e,e,x], o, V), get_count(N).`
3. Late Game:
`clear_count, minimax_value([x,o,x,o,x,o,e,e,e], x, V), get_count(N).`

Questions

1. Why is minimax optimal under the assumptions (deterministic, perfect information, zero-sum)?

Minimax is optimal under these conditions because it exhaustively searches the entire game tree and assumes the opponent plays perfectly.

- Deterministic means the outcome of every move is fully predictable (no elements of chance like dice).
- Perfect information ensures both players have complete knowledge of the current state (no hidden cards).
- Zero-sum means one player's gain is exactly the other's loss.

Because of these properties, minimax can mathematically determine the absolute worst-case scenario for any sequence of moves. By choosing the path that maximizes this worst-case outcome, it guarantees the best possible result against an opponent who is also playing perfectly.

2. What does a minimax value mean “in words”?

The minimax value of a game state is the best possible payoff a player can mathematically guarantee from that exact position, assuming that both players play optimally for the remainder of the game.

3. Why is minimax infeasible for large games like chess?

Minimax is infeasible for large games because its time complexity grows exponentially with the size of the game tree ($O(b^m)$), where b is the branching factor and m is the depth). Chess has an average branching factor of about 35 and typical games last for 80+ half-moves. This results in an estimated 10^{120} possible game states (the Shannon number). It is computationally impossible to explore a tree this vast due to limitations in processing power and memory.